

 PyMutScan

Advanced DNA Mutation Detector

 Select Reference Gene

BRCA1

 NCBI Reference Sources

Gene: BRCA1 | ID: 672

RefSeq: NM_007294.4

Location: Chr 17q21.31

Full Length: 81,189 bp

Function: DNA damage repair, tumor suppression

 [NCBI Gene Page](#) [FASTA Sequence](#) [ClinVar Variants](#) [OMIM Entry](#) Upload FASTA File

Upload .fasta or .txt

Drag and drop file here

Limit 200MB per file • FASTA, FA, TXT



PyMutScan — Advanced DNA Mutation Detector

Sequence alignment · Variant calling · Codon translation · Primer design · Drug resistance · Mutation heatmap · Patient records

 Virtual Patient Manager — Create or Load a Patient Profile [Create New Patient](#)  Load Existing Patient

Fill in patient details below. **Gene** and **Sequence** will be pulled from the current sidebar selection and query box when you click Save.

Patient Name *	Age	Gender	Blood Group
navneet	- +	M.. ▼	A+ ▼

Medical History / Symptoms

e.g. Family history of breast cancer, BRCA1 mutation 

Preliminary Diagnosis / Notes

e.g. Referred for pharmacogenomics screening 

 The current Gene (sidebar) and Query Sequence (below) will be saved with this patient profile.



Save Patient Profile



Reference

Query
Sequence

— BRCA1

(Patient)

ATGGATTATCTGCTCTTCG

67 bp demo segment · Source:
NCBI GenBank [NM_007294.4](#)

> View Full Gene
Info & Sources

Load Sample with Mutations

Paste DNA sequence or upload FASTA above (A, T, C, G only)

ATGGATTATCAGCTC
TTCGCGTTGATGAAGT
ACAAAATGTCATGAAT
GCTATGCAGAAAATCT

67 bp entered

Run Mutation Analysis

Analysis complete — 3 variant(s) detected

Total Var...	Sequenc...	Path...	Unc...	Benign
3	9...	1	1	1

- Alignment
- Mutations Table
- Sequence Stats & GC >

Pairwise Alignment

REF [BRCA1]

ATGGATTATCTGCTCTTCGCGTTGAAGAA

|||||||X|||||||X||

QRY [Patient]

ATGGATTATCAGCTCTTCGCGTTGATGA

| = match X = mismatch (mutation)



Patient Records & Database

Persistent SQLite storage — all patient profiles and analysis results are saved locally in `pymutscane_patients.db`

[Patient Profiles](#)[Analysis History](#)[Export Data](#)

No analyses saved yet. Run an analysis with an active patient to save results.